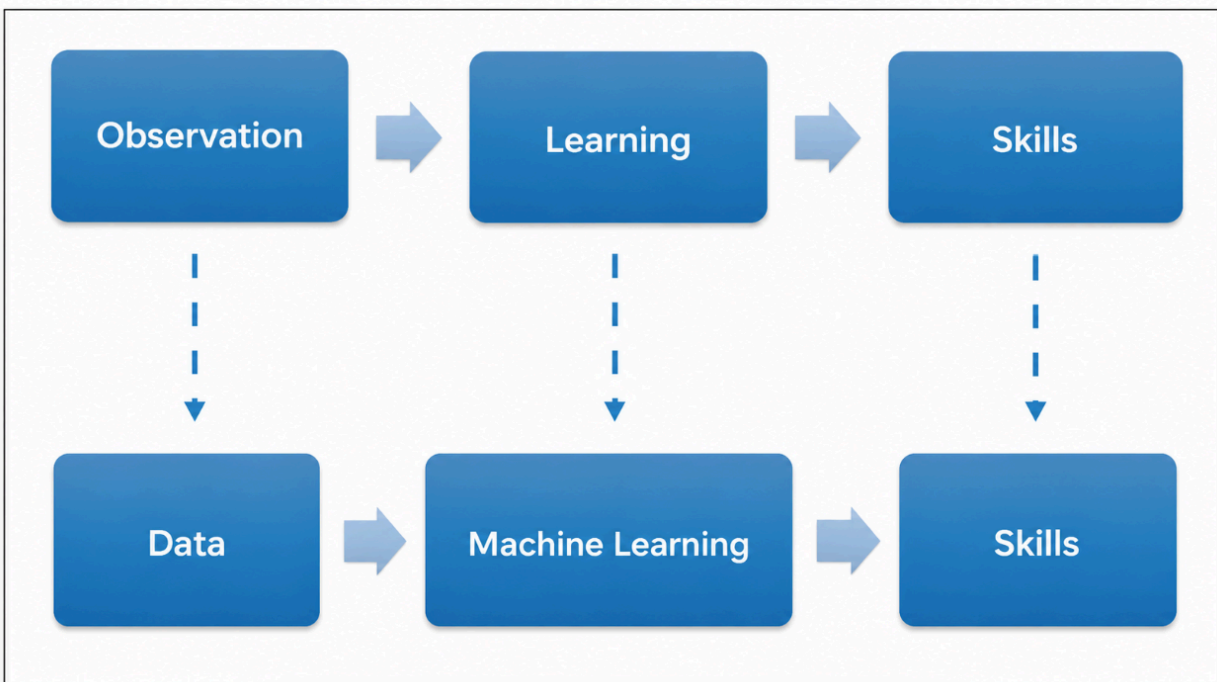


What is machine learning?

Machine learning is a core component of **artificial intelligence**, providing the essential methods that enable machines to learn and make **intelligent decisions**. It integrates knowledge from several interdisciplinary areas, including

- Probability theory.
- Statistics.
- Approximation theory.
- Convex analysis.
- Algorithm complexity theory.



At its core, machine learning works by using data to **train computational models**. These trained models can then be used to recognize patterns, solve problems, or predict future outcomes. For example, a machine learning system trained on thousands of images of cats and dogs can eventually learn to distinguish between the two automatically.

A famous real-world example is **AlphaGo**, the groundbreaking artificial intelligence system developed to play the board game Go. AlphaGo became the first AI system to defeat professional human Go players and world champions. It achieved this through a technique called **deep learning**, where the system studied enormous amounts of game

data, discovered hidden patterns and strategies, and continuously improved its gameplay through learning and experience. Rather than simply memorizing moves, AlphaGo learned the underlying principles of the game, allowing it to make intelligent decisions in complex situations.

AlphaGo Match and AI Visualizations



<https://www.youtube.com/watch?v=SUBqykXVx0A&t=38s>

Types of Machine Learning

Machine learning can generally be divided into two main categories: **supervised learning** and **unsupervised learning**. The primary difference between these approaches is whether the learning algorithm already has **labeled information** and **predefined knowledge** about the structure or categories within the dataset.

(1) Supervised Learning

Supervised learning is one of the most widely used types of machine learning. In this approach, the algorithm is trained using a **labeled dataset** (meaning that the correct answers or outcomes are already provided). The goal of the algorithm is to learn the relationship between the input data and the correct output so it can make accurate predictions when presented with new data.

Simple Teacher Example:

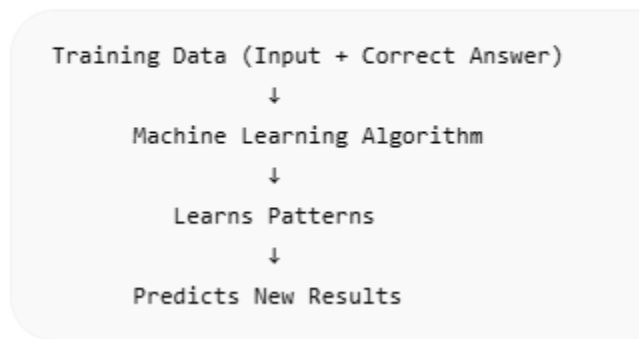
In simple terms, supervised learning is similar to learning with a teacher. The system is shown examples along with the correct answers, and through repeated training, it gradually learns how to produce the right results on its own.

How Supervised Learning Works

The process generally follows these steps:

1. Provide the algorithm with **labeled training data**.
2. The algorithm studies **patterns** and **relationships** in the data.
3. It compares its **predictions** with the correct answers.
4. Errors are corrected during training.
5. Over time, the model becomes more accurate.

Simple Visualization



Example: Image Recognition

A common example of supervised learning is image recognition.

Suppose we provide thousands of pictures of dogs to the machine, and each image is **labeled**.

Image 1 → **Dog**

Image 2 → **Dog**

Image 3 → **Dog**

These labels act as the “correct answers.” By analyzing many examples, the algorithm learns important features such as the following:

- shape
- ears
- fur patterns
- facial structure

After sufficient training, the system can examine a completely new image and determine whether it contains a dog.



Model Selection in Supervised Learning

In **supervised learning**, choosing the appropriate model is a very important step. This process is known as **model selection**. A model is essentially the mathematical or computational method used by the machine learning system to understand patterns and relationships within the data.

Different problems require different models because every dataset has unique characteristics. Therefore, selecting the right model helps improve the accuracy, efficiency, and overall performance of the machine learning system.

Why Model Selection Matters

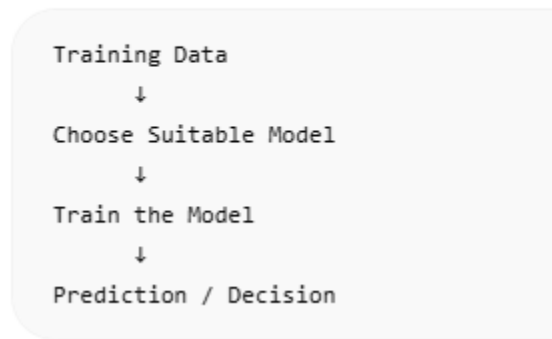
The chosen model determines:

- how the algorithm learns from data,
- how accurately it makes predictions,
- and how well it performs on **new unseen data**.

A **poor model** choice may lead to:

- inaccurate predictions,
- slow performance,
- or inability to capture important patterns in the data.

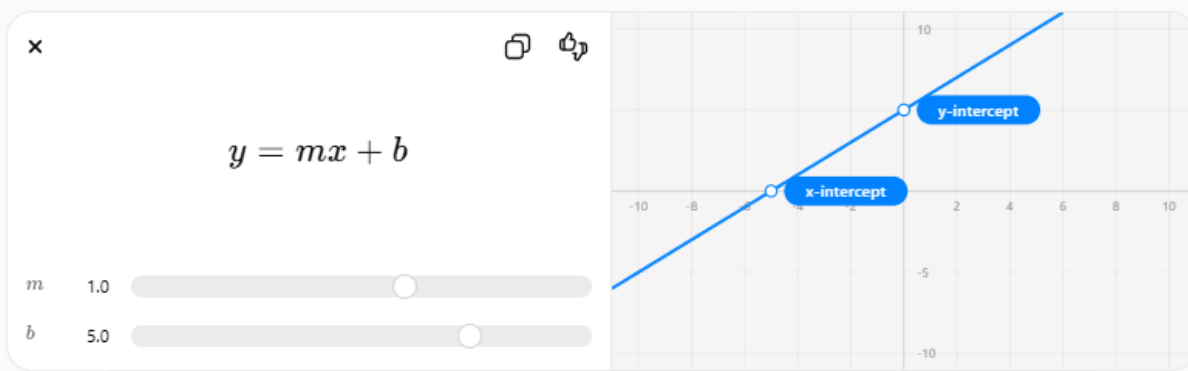
Simple Visualization



There are numerous supervised learning models such as

- **Linear regression**
- **Logistic regression**
- **Decision trees**
- **Support Vector Machine**
- **Deep neural networks**

1. Linear Regression



Linear Regression is used to predict continuous numerical values by finding a straight-line relationship between variables.

Example:

- Predicting house prices
- Predicting temperature
- Sales forecasting

Simple Idea:

As one variable changes, another variable changes proportionally.



2. Logistic Regression

Logistic Regression is mainly used for classification problems where the output belongs to categories such as:

- Yes / No
- True / False
- Spam / Not Spam

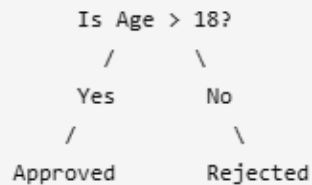
Example:

- Email spam detection
- Disease prediction
- Student pass/fail prediction

3. Decision Trees

Decision Tree Learning works like a flowchart that makes decisions step by step.

Visualization



Example:

- Loan approval systems
- Customer classification
- Medical diagnosis

Decision trees are easy to understand and visualize.

4. Support Vector Machine (SVM)

Support Vector Machine is used to separate different classes of data by finding the best boundary between them.

Example:

- Face recognition
- Text classification
- Image classification

Main Idea:

Find the optimal line or boundary that separates categories.

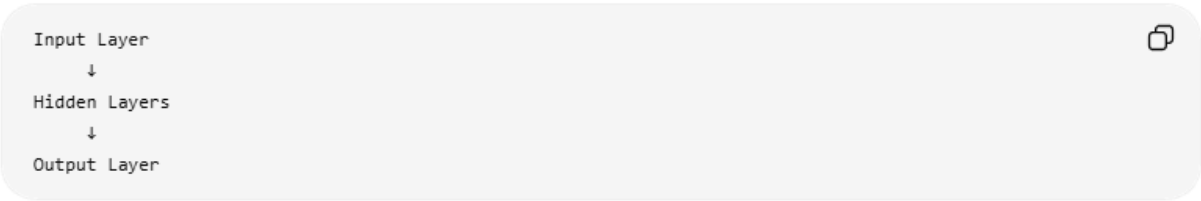
5. Deep Neural Networks

Deep Learning uses layers of artificial neurons inspired by the human brain to learn highly complex patterns from large datasets.

Example:

- Self-driving cars
- Speech recognition
- Chatbots
- Image recognition
- AI assistants

Visualization



Deep neural networks are especially powerful for large-scale AI applications.

Choosing the Right Model

The best model depends on several factors:

Factor	Explanation
Type of data	Numerical, text, images, audio
Problem type	Classification or prediction
Dataset size	Small or large
Complexity	Simple or complex patterns
Accuracy needs	Basic or high-performance

Key Concept

- There is no single “best” model for every problem.
- Different problems require different models.

- A machine learning engineer selects the model that best fits the dataset and the task being solved.

Summary

Model selection is a critical step in supervised learning because it determines how effectively the system can learn from data and make predictions. Common supervised learning models include:

- Linear Regression
- Logistic Regression
- Decision Trees
- Support Vector Machines (SVM)
- Deep Neural Networks

The final choice depends on the nature of the data and the specific problem being addressed.

Feature Engineering

Feature engineering is the process of preparing raw data so a model can effectively use it. This includes cleaning the data, handling missing values, and transforming it into a better format using techniques such as normalization or standardization. It may also involve selecting the most important features and creating new ones. Good feature engineering can greatly improve how well a model performs.

Training and Optimization

In this step, we use labeled data to teach the model how to make predictions. This is done by defining a loss function that measures errors and then using optimization methods to reduce those errors. The model adjusts its internal parameters step by step to improve accuracy. Common optimization techniques include **gradient descent** and **stochastic gradient descent**.

Model Evaluation

After training the model, we test how well it works on new, unseen data. This helps us understand its real-world performance. We use evaluation metrics such as accuracy, precision, recall, F1 score, and ROC curve to measure how good the model is. These results show whether the model is reliable and suitable for practical use.

Summary

Supervised learning is a method where a **model is trained using labeled data** to make predictions on **new data**. The main steps include choosing a model, preparing and improving the data (feature engineering), training and optimizing the model, and finally evaluating its performance. Together, these steps form the core process of supervised learning.

(2) Unsupervised Learning

Unsupervised learning is a type of machine learning in which an algorithm is trained using data that has **no labels or predefined answers**. Instead of being told what each data item represents, the system must identify patterns, similarities, and relationships on its own.

For example, in image classification, a large collection of cat and dog images can be provided without any labels. By analyzing visual features and similarities within the images, the algorithm can automatically organize them into separate groups, such as cat images and dog images.

Unsupervised learning helps machines discover **hidden patterns** and **group similar data** without using labeled examples.

Unsupervised Learning

